

Education

McMaster University – B.Eng. BME, Mechanical and Biomedical Engineering. Dean's Honor List.

April 2026

- **Relevant courses:** Biomechanics, Computational Biomechanics, CAD

Experience

Mechanical Design Engineer Co-op – Enbridge Gas – Chatham, ON

May 2024 – Aug 2025

- Signed as Mechanical Engineering Lead on an \$8M gas station project and supported 3 additional concurrent projects, reviewing third-party design deliverables (P&IDs, drawings, equipment lists) and managing timelines and budget
- Signed as Mechanical Lead on an \$8M expansion project, managed budgets, timelines, and 3 concurrent projects
- Reviewed third-party design packages (P&IDs, drawings, pipe size calculations) for regulatory compliance, alignment with Enbridge standards and mandated typical drawings to optimize design as needed
- Conducted field investigations, including on-site visits, reviewing site photographs, plans, and Navisworks models against Enbridge, safety, and MTO/TAC requirements (station layouts, equipment clearances, entrances, driveways)
- Gained working knowledge of field safety-testing equipment and protocols through BlueBeam design review meetings
- Collaborated in interdisciplinary HAZOP risk assessments on Bluebeam Revu with external and internal stakeholders
- Investigated a safety issue involving an incorrectly installed U-bolt, identifying missing detail in the engineering drawing
- Participated in RCA on a Pressure Relief Valve (PRV) field event, identifying drawing detail gaps in a misinstalled U-bolt
- Reviewed and compared 200+ legacy engineering standards and typical drawings following a company merger, flagged outdated or duplicate versions for SME review and final selection
- Performed gap analysis across 200+ post-merger legacy standards and collaborated with SMEs to create a masterlist
- Developed a VBA tool to locate missing MTRs across 1,000+ improperly filed POs; saved 50+ engineering hours
- Built and deployed an Excel project tracking tool across 10 engineering teams (100+ users) to monitor deliverables progress; presented to leadership in Calgary, securing approval to formalize the tool into engineering procedure
- Signed as Mechanical Engineering Lead on an \$8M gas station project and supported 3 additional concurrent projects, reviewing third-party design deliverables (P&IDs, drawings, equipment lists) on Bluebeam Revu
- Built and deployed an Excel-based project tracking dashboard across 10 engineering teams (100+ users) to track engineering and design deliverables against 30/60/90% project milestones, quantifying and displaying progress
- Structured the tool with dedicated sections for action items, outstanding deliverables, and overall package completeness to monitor project performance and guide meeting readiness
- Presented the tool to leadership in Calgary, where it was selected to replace the project controls team's existing tool; trained engineering teams on how to use it and secured approval to formalize it into engineering procedure
- Organized equipment lists structure for maintenance teams to streamline uploads into CMMS software (Maximo, Oracle)
- Supported Kahua project management software rollout by mapping user workflows and providing hands-on training

Mechanical Engineering Intern – ReFem Tech Startup – Toronto, ON

Sept 2025 – June 2026

- Designed a dual gel-suction IUD insertion device in SolidWorks, consolidating three instruments into one system
- Designed an IUD insertion device in SolidWorks, consolidating 3 tools into one; created engineering drawings
- Conducted prototype testing across five iterations, 3D-printing and diagnosing mechanical failures, including tolerance, sizing, and ergonomic issues, and implementing corrective design changes
- 3D-printed 5 iterations to test functionality, including a piping redesign to route around internal electrical sensors
Authored a design specification sheet documenting how the device met regulatory standards (ISO 10993, ISO 14971)
- Co-authored the Technical Specifications Document to drive design decisions for an FDA 510(k) submission pathway
- Conducted FMEA risk assessments and researched ISO 10993/14971 standards to ensure full regulatory compliance
Interviewed 3 clinicians to shape the design direction; helped secure \$10,000+ in startup pitch awards
- Contributed to business development: tracked project costs and budget, identified and pursued grant and competition funding opportunities, and conducted financial and market analysis, helping secure \$10,000+ in startup pitch awards
- Designed an electromechanical IUD insertion device in SolidWorks, consolidating three instruments into one system
- Coordinated weekly update meetings with 5 team members, tracking project milestones in a Gantt chart

- Sourced and scheduled interviews with clinicians for design input, leading meetings, taking notes, and follow-up
- Tracked design progress across 5 prototype iterations, documenting mechanical failures and corrective changes

Mechanical Engineering Research Assistant Co-op – McMaster University – Hamilton, ON May 2021 - Aug 2021

- Ran wet-lab experiments and COMSOL simulations to evaluate flow uniformity in a chromatography device for vaccine purification; diagnosed leaks, resolving them via added fasteners and a gasket, and presented findings to faculty
- Ran lab experiments and COMSOL simulations to evaluate flow uniformity in a chromatography device for vaccine purification; diagnosed leaks and resolved them by increasing fastener count and introducing a gasket to the design
- Designed test procedures, documented findings in a technical report, and presented results to the research group

Selected Projects

Biofill: Syringe-Filling Device (Patent Pending, Inventor) Capstone

- Designed an automated bioink syringe-filling device for Tessella Biosciences, now adopted into their workflow, filling 2–10 mL syringes (± 0.05 mL accuracy) in a biosafety cabinet (BSC) using 95% 3D-printed components
- Investigated mechanical failures, including piston detachment and locking-mechanism shear, using FEA, physical testing, and mechanical analysis to identify failure mechanisms and validate corrective designs
- Met with the stakeholder weekly to track feedback and managed a Gantt chart to report on design progress
- Investigated mechanical failures, including piston detachment and locking-mechanism shear, using FEA, physical testing, and mechanical analysis to identify failure mechanisms and create component maintenance plans
- Ran FEA simulations in Autodesk Inventor and physical testing on the device housing to validate it could withstand motor force and piston loading, guarding against deformation, fastener shear, and projectile risk
- Authored a technical product report documenting design specifications, testing and failures for patent application
- Authored the invention disclosure and compiled engineering drawings, BOM, technical specifications, design iteration history, and testing data to support a patent application
- Co-authored the invention disclosure and compiled engineering drawings, BOM, spare parts list, technical specifications, design iteration history, and testing data to support manufacturing and formal patent application
- Programmed Arduino-based control logic in C++ to automate the fill cycle and limit dead volume to 1mL per cycle

Pediatric Cerebral Palsy Remote Gait Analysis Device Course Project

- Measured ankle dorsiflexion using gyroscope/pressure sensors to analyze kinematic data associated with cerebral palsy.

Computational Human Movement Analysis Course Project

- Used OpenSim to track human kinematics, applying biomechanics to analyze trajectory, velocity, and movement data.

Pickup Truck CAD & Simulation Course Project

- Designed pickup truck assembly in Inventor (drivetrain, chassis, mounts) and created a full engineering drawing package.
- Performed stress, motion, and CAM simulations to evaluate structural integrity and mechanical behavior under load.

Extracurricular Experience

Tax Volunteer - Canadian Volunteer Income Tax Program (CVITP) Sep 2025 – Present

- Filed 100+ tax returns for low-income clients at free community clinics with professionalism and confidentiality.

Skills

- **Field & Testing:** Field Investigations, Site Surveys, Failure Analysis, Kinematic Data Processing, Risk Assessment (FMEA)
- **Software & Analysis:** SolidWorks, Autodesk Inventor, Navisworks, Bluebeam, COMSOL, OpenSim, Python, Excel (VBA)
- **Certifications & Licenses:** Ontario Class G Driver's License, EIT Eligible (PEO)